

Mason Canfield

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Education

James Madison University

B.S. Engineering (Electro-Mechanical Systems)

Minors: Mathematics, Robotics

GPA: 3.54 | May 2027

Experience

Engineering Intern

Mohawk Industries | May 2026–Aug 2026

- Modeled carpet roll length as a function of face weight using log-linear regression in Minitab, replacing a flat feel-based loading target with a style-specific one; validated with VIF diagnostics, a Chow test, and ANOVA/Tukey HSD tiering on 500+ self-collected rolls across 89 styles
- Improved predicted R^2 from 73.3% to 85.6% over the legacy equation, cutting roll counts on six flagship styles from 6,411 to 5,615 a year (6–13.6% plant-wide) and identifying **\$44.7K–\$65.3K** in annual savings
- Partnered with Maintenance to apply Fan Affinity Laws, quantifying energy savings from reduced-speed operation of a spinning supply fan via VFD; delivered a validated savings case of **9.9 tons of CO₂, 37,800 kWh, and \$4,100** per year
- Digitized and modernized 100+ lockout/tagout procedures facility-wide in BradyLink for compliance and standardization

Student-Athlete Tutor (Math & Science)

JMU | Jan 2025–Present

- Applied structured problem-solving methods to explain calculus and physics concepts
- Created step-by-step solution frameworks to improve student performance and retention

Landscaping Technician

Steeplechase Lawn Services | May 2023–Aug 2025

- Maintained, repaired, and troubleshoot mechanical equipment and small engines in the field
- Delivered dependable, on-schedule services across residential and commercial properties

Projects

EarlyEar — IoT Livestock Health Monitor

Independent Project

- Designed and built a BLE ear tag for early bovine respiratory disease detection in show cattle, integrating medical-grade infrared thermometry (0.2°C accuracy), MEMS accelerometer, humidity sensing, and LiPo fuel gauging on an ESP32-S3
- Cut measured system current **30x**, from 90 mA to 3 mA, through deep-sleep duty cycling and low-power component selection, extending projected field life from under 4 hours to 4 days; verified with shunt/DMM instrumentation
- Replaced local OLED readout with a wireless telemetry pipeline: a fixed 22-byte binary packet with a version byte and CRC check, a Python host decoder producing analysis-ready CSV, and on-tag computation of activity and heat-stress values

Shell Eco-Marathon — Frame & Data Acquisition Lead

JMU

- Built a physics-based MATLAB/Simulink/Stateflow road-load model across nine iterations predicting lap-by-lap fuel economy over the Indianapolis infield course from GPS-derived grade and curvature, using a burn-coast control strategy, a dyno-calibrated torque map, and a compliance check
- Ran five-parameter sweeps that ranked aerodynamic drag as roughly 3x more influential than mass per realistic design increment and set the team's build priorities
- Reduce chassis mass while preserving torsional stiffness, using FEA to target material removal and dial-gauges to verify predicted stiffness on the built frame
- Design, test, and prototype a composite aerodynamic shell to replace the vehicle's rigid plexiglass panel body, targeting the drag reduction identified as the highest-value change by the sensitivity study
- Specify and integrate the vehicle data acquisition system – thermocouples, hall-effect wheel speed, GPS, and engine feedback – to deliver live driver feedback and the measured inputs needed to calibrate the simulation

Activities

Madison Aerospace Club

JMU | Oct 2025–Present

- Conducted trade studies on 4+ fin geometries, motor selections, and airframe configurations using OpenRocket, evaluating stability margin and apogee sensitivity
- Completed Level 1 High Power Rocketry Launch: selected the Aerotech H169WS-13A motor and trimmed fin span from 4 in to 3 in for a 120 ft (3.8%) apogee gain at a 2.1-cal stability margin; launched and recovered successfully, April 2026

Additional: Madison Welders Club (Spring 2026–Present); Intramural Soccer & Flag Football (Fall 2023–Present)

Skills

CAD & Design: SOLIDWORKS (CSWA), Fusion 360, parametric modeling, technical drawings, GD&T basics

Analysis & Simulation: ANSYS Mechanical (FEA), MATLAB, Simulink, Stateflow, OpenRocket

Embedded & Hardware: ESP32, Arduino, Embedded C++, BLE, sensor integration, LiPo power systems

Fabrication & Testing: 3D printing (PETG/FDM), soldering, composite layup

Data and Statistical Analysis: Excel, Minitab, regression modeling, ANOVA, VIF diagnostics, Python (basic)